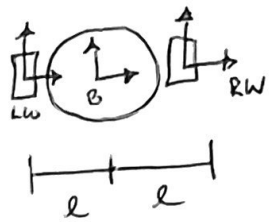
Turtle Bot Kinematics Equations

Diagram:



Body Twist in Wheel Frames:

Adjoints:

$$A_{B_RW} = \begin{bmatrix} 1 & 0 & 0 \\ l & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_{RW_B} = \begin{bmatrix} 1 & 0 & 0 \\ -l & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A_{B_LW} = \begin{bmatrix} 1 & 0 & 0 \\ -l & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_{LW_B} = \begin{bmatrix} 1 & 0 & 0 \\ l & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

* wheel constraint

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = \begin{bmatrix} r\dot{\phi}_i \\ 0 \end{bmatrix}$$

* Twist Formula: $\mathcal{V}_i = A_{ib}\mathcal{V}_b$

RW_Body Twist: $V_{B_RW} = A_{RW_B} \cdot V_B = \begin{bmatrix} 1 & 0 & 0 \\ -l & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -l\dot{\theta} + \dot{x} \\ \dot{y} \end{bmatrix}$

lw_Body Twist: $V_{B_LW} = A_{LW_B} \cdot V_B = \begin{bmatrix} 1 & 0 & 0 \\ l & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ l\dot{\theta} + \dot{x} \\ \dot{y} \end{bmatrix}$

Wheel Equations of Motion:

RW: $\begin{bmatrix} \dot{\theta} \\ r\dot{\phi}_{rw} \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -l\dot{\theta} + \dot{x} \\ \dot{y} \end{bmatrix} \rightarrow \begin{bmatrix} \dot{\phi}_{RW} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{-l}{r} & 1/r & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ v_x \\ v_y \end{bmatrix}$

LW: $\begin{bmatrix} \dot{\theta} \\ r\dot{\phi}_{LW} \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ l\dot{\theta} + \dot{x} \\ \dot{y} \end{bmatrix} \rightarrow \begin{bmatrix} \dot{\phi}_{LW} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{l}{r} & 1/r & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ v_x \\ v_y \end{bmatrix}$

Inverse Kinematics:

$\dot{\phi} = H V_B$, (Relation of body twists to wheel velocities)

$$\begin{bmatrix} \dot{\phi}_{RW} \\ \dot{\phi}_{LW} \end{bmatrix} = \frac{1}{r} \underbrace{\begin{bmatrix} -\ell & 1 & 0 \\ \ell & 1 & 0 \end{bmatrix}}_{H} \begin{bmatrix} \dot{\theta} \\ v_x \\ v_y \end{bmatrix}$$

$0 = Q V_B$, (sideways motion constraints)

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{Q} \begin{bmatrix} \dot{\theta} \\ v_x \\ v_y \end{bmatrix}$$

Forward Kinematics:

$$V_B = H^{+} u = H^{T}(HH^{T})^{-1} \Rightarrow V_B = \frac{r}{2} \begin{bmatrix} -1/\ell & 1/\ell \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\phi}_{RW} \\ \dot{\phi}_{LW} \end{bmatrix}$$

$$\begin{bmatrix} \dot{\theta} \\ v_x \\ v_y \end{bmatrix} = \begin{bmatrix} -r\dot{\phi}_{RW}/2\ell + r\dot{\phi}_{LW}/2\ell \\ r\dot{\phi}_{RW}/2 + r\dot{\phi}_{LW}/2 \\ 0 + 0 \end{bmatrix}$$

① $\dot{\theta} = -\frac{r\dot{\phi}_{RW}}{2l} + \frac{r\dot{\phi}_{LW}}{2l} = \frac{r}{2l}\left(\dot{\phi}_{LW} - \dot{\phi}_{RW}\right)$

↳ wheel velocities.

② $V_x = r/2\left(\dot{\phi}_{RW} + \dot{\phi}_{LW}\right)$

$l = \frac{\text{wheel_track}}{2}$

$r =$ wheel_radius

⑤ $\dot{\phi}_{rw} = -\frac{l}{r}\dot{\theta} + \frac{V_x}{r}$

⑥ $\dot{\phi}_{LW} = \frac{l}{r}\dot{\theta} + \frac{V_x}{r}$

linear

③ $-l\dot{\theta} + V_x$

④ $l\dot{\theta} + V_x$